

Assignment-1

①

a) exponents = $\{-2, -1, 0, 1, 2\}$
 $m = 4$

Standard

$$0. d_1 d_2 d_3 d_4 \times \beta^e, \quad d_1 = 1$$

w/o -ve, $\min = 0.1000 \times 2^{-2}$
 $\max = 0.1111 \times 2^2$

with -ve
 $\min = -0.1111 \times 2^2$
 $\max = 0.1111 \times 2^2$

Normalized

$$0.1 d_1 d_2 d_3 d_4 \times \beta^e,$$

w/o -ve
 $\min = 0.10000 \times 2^{-2}$
 $\max = 0.11111 \times 2^2$

with -ve,
 $\min = -0.11111 \times 2^2$
 $\max = 0.11111 \times 2^2$

Denormalized

$$1. d_1 d_2 d_3 d_4 \times \beta^e$$

w/o -ve,
 $\min = 1.0000 \times 2^{-2}$
 $\max = 1.1111 \times 2^2$

with -ve,
 $\min = -1.1111 \times 2^2$
 $\max = 1.1111 \times 2^2$

b) Standard $\rightarrow$ 5 expts
 w/o -ve $0.1 \overset{d_1}{\square} \overset{d_2}{\square} \overset{d_3}{\square} \overset{d_4}{\square} \times \beta^e \therefore 2^3 \times 5 = 40$
 $0/1 \ 0/1 \ 0/1$

with -ve, $40 \times 2 = 80$

Normalized $\rightarrow$
 w/o -ve $0.1 \overset{d_1}{\square} \overset{d_2}{\square} \overset{d_3}{\square} \overset{d_4}{\square} \times \beta^e \therefore 2^4 \times 5 = 80$

with -ve, $80 \times 2 = 160$

Denormalized $\rightarrow$
 w/o -ve, $1. \square \square \square \square \times \beta^e \therefore 2^4 \times 5 = 80$

with -ve, $80 \times 2 = 160$

c)

0.1000×2^{-2}	0.1000×2^{-1}	Try yourself	0.1000×2^2
0.1001×2^{-2}	0.1001×2^{-1}		0.1001×2^2
$\vdots$	$\vdots$		$\vdots$
$\vdots$	$\vdots$		$\vdots$
0.1111×2^{-2}	0.1111×2^{-1}		0.1111×2^2

②

(a)

$$x^2 - 10x + 3 = 0 \Rightarrow x_1 = 5 - \sqrt{22} = 0.3096$$

$$x_1 = 5 + \sqrt{22} = 9.690$$

(b)

$$\begin{aligned}\sqrt{22} &= 4.690415 \\ &= 4.690 \quad [\text{upto 4 a.f.}]\end{aligned}$$

$$\therefore x_1 = 5 + \sqrt{22} = 9.690 \quad \checkmark$$

$$x_2 = 5 - \sqrt{22} = 5 - 4.69 = \boxed{0.31}$$

Loss of significance
for subtracting 2
nearly equal numbers.

(c)

solution,

$$x^2 - 10x + 3 = 0 \Rightarrow x^2 - (a+b)x + ab = 0$$

$$a + b = 10$$

$$ab = 3$$

$$\therefore b = \frac{3}{a}$$

$$\text{If } a = 9.690, \quad b = \frac{3}{9.690}$$

$$= 0.3096$$

(12)

(a)

7)

$$x = \frac{3}{8}$$

$$= \frac{(11)_2}{2^3}$$

$$= 0.011$$

$$= (0.11)_2 \times 2^{-1}$$

$$y = \frac{7}{8}$$

$$= (0.111)_2 \times 2^0$$

$$A, m=4$$

$$f(x) = x$$

$$f(y) = y$$

$$x \times y = f(x) \times f(y)$$

$$= \frac{(0.11)_2 \times 2^{-1}}{\times \frac{(0.111)_2 \times 2^0}{}}$$

$$= \frac{3}{8} \times \frac{7}{8}$$

$$= \frac{21}{64}$$

$$= (0.010101)_2$$

$$= (0.10101)_2 \times 2^{-1}$$

$$f(x \times y) = (0.1010 \mid 1)_2 \times 2^{-1}$$

$$= (0.1010)_2 \times 2^{-1}$$

$$= \frac{5}{8} \times \frac{1}{2} = \frac{5}{16} = \frac{20}{64}$$

$$\therefore \text{Loss of significance} = \frac{41}{64} - \frac{20}{64} \\ = \frac{7}{64}$$

$\therefore$ Can be stored, as exponent not exceeding -2 on the left and 2 on the right, but will lose significance

④

(a)

$$x = (6.325)_{10}$$

$$= 110.0101001\dots$$

$$= (0.1100101001)_{2} \dots \times 2^3$$

(b)

$$m = 3$$

$$f(x) = (0.110 \mid 0101001\dots)_2 \times 2^3$$

$$= (0.110)_2 \times 2^3$$

(c)

~ ~ ~ 3

$$f(x) = (0.110)_2 \times 2^1$$

$$= \frac{6}{8} \times 8$$

$$= 6$$

$$\therefore \text{Rounding error} = \frac{|f(x) - x|}{x} \times 100\%$$

$$= \frac{|6 - 6.325|}{6.325} \times 100\%$$

$$= 5.14\%$$

$$E_m = \frac{1}{2} \beta^{1-m}, \text{ for standard}$$

$$= \frac{1}{2} \times 2^{1-3}$$

$$= \frac{1}{2} \times 2^{-2}$$

$$= \frac{1}{8}$$

Assignment-2

①

(a)

.....

3 points $\rightarrow$ $\angle$ degree

$$P(x) = a_0 + a_1x + a_2x^2$$

$$P(2) = 10$$

$$P(4) = 20$$

$$\Rightarrow a_0 + 2a_1 + 4a_2 = 10 \quad \Rightarrow a_0 + 4a_1 + 16a_2 = 20$$

$$P(6) = 25$$

$$\Rightarrow a_0 + 6a_1 + 36a_2 = 25$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 6 & 36 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = V^{-1} Y \quad \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -5 \\ \frac{35}{4} \\ -\frac{5}{8} \end{bmatrix}$$

$$v(t) = -5 + \frac{35}{4}t - \frac{5}{8}t^2$$

$$\therefore a(t) = v'(t) = \frac{35}{4} - \frac{5}{4}t$$

$$a(7) = \frac{35}{4} - \frac{35}{4} = 0 \text{ m s}^{-2}$$

$$\underline{t \quad v(t) \quad \underline{\underline{(b)}}$$

t_0	2	10
t_1	4	20
t_2	6	25

$$P(t) = l_0(t) f(t_0) + l_1(t) f(t_1) + l_2(t) f(t_2)$$

$$l_0(t) = \frac{t-t_1}{t_0-t_1} \cdot \frac{t_1-t_2}{t_0-t_2} = \frac{(t-4)(t-6)}{-2 \times -4} \\ = \frac{1}{8} (t-4)(t-6)$$

$$l_1(t) = \frac{t-t_0}{t_1-t_0} \cdot \frac{t-t_2}{t_1-t_2} = \frac{1}{-4} (t-2)(t-6)$$

$$l_2(t) = \frac{t-t_0}{t_2-t_0} \cdot \frac{t-t_1}{t_2-t_1} = \frac{1}{8} (t-2)(t-4)$$

$$x(t) = \frac{1}{8} (t-4)(t-6) \times 10 + \frac{(-\frac{1}{4})}{20} (t-2)(t-6) \times 20 \\ + \frac{1}{8} (t-2)(t-4) \times 25$$

Newton's method ^(c). Degree = 3.

②

x	y
$-\frac{\pi}{2}$	$\frac{\pi}{2}$

$$\begin{array}{c} 0 \\ \frac{\pi}{2} \end{array} \left| \begin{array}{c} 0 \\ \frac{\pi}{2} \end{array} \right.$$

$$p(x) = f[x_0] + (x-x_0) f[x_0, x_1] + (x-x_0)(x-x_1) f[x_0, x_1, x_2]$$

$$\begin{array}{l} x_0 = -\frac{\pi}{2} \quad f[x_0] = \frac{\pi}{2} \\ x_1 = 0 \quad f[x_1] = 0 \\ x_2 = \frac{\pi}{2} \quad f[x_2] = \frac{\pi}{2} \end{array} \left. \begin{array}{l} \rightarrow f[x_0, x_1] = \frac{0 - \frac{\pi}{2}}{0 - (-\frac{\pi}{2})} = -1 \\ \rightarrow f[x_1, x_2] = \frac{\frac{\pi}{2} - 0}{\frac{\pi}{2} - 0} = 1 \end{array} \right\} f[x_0, x_1, x_2] = \frac{1 - (-1)}{\frac{\pi}{2} - (-\frac{\pi}{2})} = \frac{2}{\pi}$$

$$p(x) = \frac{\pi}{2} + (x + \frac{\pi}{2})(-1) + (x + \frac{\pi}{2})(x - 0) \cdot \frac{2}{\pi}$$

③

(a)

$$x_0 = 2$$

we need to find $f'(2)$

B.D

$$f'(2) = \frac{f(2) - f(2-h)}{h}$$

C.D.

$$f'(2) = \frac{f(2+h) - f(2-h)}{2h}$$

h	B.D	C.D

0.1	-2.5972	-2.6464
0.01	-2.6459	-2.6508
0.001	-2.6504	-2.6508

(b)

F.D

$$\text{Max error} = \frac{|f''(\xi)|}{2!} \times h$$

$$= \frac{|2\cos x - 2\sin x|}{2} \times h$$

$$|2\cos x - 2\sin x| \leq |2\cos x| + |2\sin x|$$

$$\leq |2\cos x| + |2\sin x|$$

Since $x_0 = 2$, range will be $[2, 2+h]$
 $= [2, 2.1]$

$$\therefore \leq |2\cos x| + |2\sin x|$$

$$\leq 2 \times \cos 2 + 2 \sin 2.1$$

$$\leq 0.88$$

$$\dots \dots 0.88 \times 0.1 = 0.088$$

$$\text{Max error} = \frac{1}{2} h^2$$

C.D Range is $[2-h, 2+h]$
 $= [1.9, 2.1]$

$$\frac{|f'''(\xi)|}{3!} h^2 = \frac{|-2\sin x - 2\cos x|}{3!} \times h^2$$

$$= \frac{|2\sin x + 2\cos x|}{3!} \times h^2$$

Max error

$$\leq |2\sin x| + |2\cos x|$$

$$\leq 2\sin 2.1 + 2\cos 1.9$$

$$\leq 1.0798$$

$$\text{Max error} \leq \frac{1.0798}{6} \times 0.1$$

$$= 0.0179$$

$n = 2$ [3 points] (c)

$$\text{Error} = \frac{|f'''(\xi)|}{3!} \times (x-x_0)(x-x_1)(x-x_2)$$

$$= \sqrt{\text{From (b)}}$$

$$= \frac{|2\sin x + 2\cos x|}{6} \times \frac{(\pi^3 - \frac{\pi^2}{16}x)}{1}$$



$$W'(x) = 0$$

$$\Rightarrow 3x^2 - \frac{\pi^2}{16} = 0$$

$$\Rightarrow x = \pm \frac{\pi}{4\sqrt{3}}$$

$$\text{Range} = \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$$

$$\leq |2\sin x| + |2\cos x|$$

$$\leq 2\sin \frac{\pi}{4} + 2\cos 0$$

$$\leq 3.414$$

x	$w(x)$
$-\frac{\pi}{4}$	0
$-\frac{\pi}{4\sqrt{3}}$	0.1864 ✓ max
$\frac{\pi}{4\sqrt{3}}$	0
$\frac{\pi}{4}$	-0.1864

$$\therefore Am = \frac{3.414}{6} \times 0.1864$$

④

(a)

Chebyshev nodes

(b)

$$\text{center} = 0 \quad n = 2$$

$$j = 0, 1, 2, 3$$

$$x_0 = 0 + 2 \cos \left\{ \frac{2 \times 0 + 1}{2(3+1)} \pi \right\}$$

$$x_1 = 0 + 2 \cos \left\{ \frac{2 \times 1 + 1}{2(3+1)} \pi \right\}$$

$$x_2 = 0 + 2 \cos \left\{ \frac{2 \times 2 + 1}{2(3+1)} \pi \right\}$$

$$x_3 = 0 + 2 \cos \left\{ \frac{2 \times 3 + 1}{2(3+1)} \pi \right\}$$

(c)

$$n = 1 \text{ [2 points]}$$

$$P_3(x) = f(x_0) h_0(x) + f(x_1) h_1(x) + f'(x_0) \hat{h}_0(x) + f'(x_1) \hat{h}_1(x)$$

	x	$f(x)$	$f'(x)$
x_0	0	1	0
x_1	π	-1	-1

$$x_1 \quad \overline{\Sigma} \quad \downarrow$$

$$p_3(x) = f(x_0)h_0(x) + f'(x_1)\hat{h}_1(x)$$